

Portfolio

Data Analyst Internship – Jabatan Pendidikan Negeri Sembilan (JPNS)

Project: War Room SPM 2023

Objective: Designed to assist education officers in planning strategic solutions for specific subjects and identifying key performance targets for upcoming SPM exam results based on historical trends.

Scope & Technical Stack:

- **Database:** Utilized Google Sheets as the primary engine for data storage and management.
- **Frontend Visualization:** Built interactive dashboards in Looker Studio to provide real-time insights to stakeholders.
- **Security:** Implemented AppSheet to manage secure user authentication and granular authorization levels.
- **Data Analysis:** Conducted comprehensive exam result analysis, including longitudinal data from past SPM results (2017-2022) and internal school assessments (Mid and Final year exams).

Final Year Project – Personalized Investment Platform Recommendation System

Project: Sentiment-Driven Digital Asset Advisor

Objective: Engineered a dynamic web application to assist individual investors in selecting digital financial services by prioritizing authentic user experiences over marketing claims or generic numerical ratings.

Scope & Technical Stack:

- **Machine Learning & NLP:** Leveraged BERT-base architecture, Aspect-Based Sentiment Analysis (ABSA), PyTorch, and HuggingFace for advanced text interpretation.
- **Backend & Processing:** Utilized Python, Flask, Pandas, and NumPy for robust data manipulation and server-side logic.

- **Data Acquisition:** Automated large-scale retrieval of 8,000+ reviews via google-play-scraper and custom HTML parsing scripts.
- **Frontend Visualization:** Built a responsive UI using HTML, CSS, and JavaScript, incorporating Chart.js for comparative visual analytics.

Key Contributions & Architecture:

- **NLP Modeling:** Fine-tuned BERT to perform simultaneous aspect extraction and polarity classification across key dimensions like trust, usability, and fees.
- **Recommendation Engine:** Designed an algorithmic scoring system that computes personalized rankings by weighting sentiment data against user-defined priorities.
- **Full-Stack Development:** Deployed a Flask application featuring real-time review processing and interactive leaderboards with donut and radar chart visualizations.
- **Performance Validation:** Achieved a 99% prediction accuracy and F1-score, alongside an outstanding System Usability Scale (SUS) score of 91 during testing.

Project Outcomes:

- The ABSA model demonstrated exceptional precision in classifying sentiment across positive, neutral, and negative categories.
- The finalized platform provided a highly intuitive stakeholder experience, confirming the effectiveness of the sentiment-driven approach.

Full-Stack Development – Elvira Admin Portal

Project: Integrated Workforce Management Ecosystem

Objective: Engineered a centralized digital workspace to optimize internal operations by consolidating disparate administrative workflows into a unified, high-efficiency platform.

Scope & Technical Stack:

- **Framework & Language:** Developed using Laravel and PHP for scalable server-side architecture.
- **Database:** Implemented MySQL for robust relational data management and storage.
- **Frontend & UI:** Crafted responsive interfaces using Blade Templates, Tailwind CSS, Alpine.js, and Vite.
- **Security:** Configured sophisticated role-based access controls and secure authentication protocols.

Key Contributions & Architecture:

- **System Design:** Architected the complete application structure leveraging the Laravel MVC design pattern.
- **Module Engineering:** Built core business engines for attendance tracking, leave processing, and expense claims.
- **Resource Management:** Developed comprehensive systems for inventory monitoring, document repositories, and task oversight.

- **Workflow Automation:** Programmed complex validation logic and multi-tier approval pipelines for operational requests.
- **Data Visualization:** Designed dynamic dashboards providing role-specific insights and real-time activity monitoring.

Project Outcomes:

- Achieved total digitization of administrative records, significantly reducing manual data handling and redundancy.
- Enhanced organizational transparency and accountability through improved visibility of staff activities and requests.
- Optimized inter-departmental collaboration by providing a centralized communication and document management hub.